

Quantics Tensor-Train Finite-Volume Solver for the Convecting Taylor–Green Vortex

Reynolds scaling of memory, error and time-to-solution in tensor-train form, with a hardware-native pathway flown at 42 qubits

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-airbus-gic2026>

Summary

92,183×

less memory than the dense velocity field at $Re = 10^4$ on a $4,096^2$ grid — the field held at bond dimension 3 with L2 error 4.4×10^{-10} against the analytic solution, in the tensor-network-within-finite-volume track this challenge explicitly invites

$10^{-8} - 10^{-10}$

L2 error across three decades of Reynolds number (Re 10, 100, 10,000) with rank ≤ 8 , cell averages and face fluxes as the statement's finite-volume framework requires; QTT and dense stencils agree to 10^{-10}

42 qubits

lattice-gas medium with a Taylor–Green parity preparation on IBM Heron, six pre-registered gates passed on its operator identity and waveform — the dynamical primitive a gate-model fluid solver builds on

Why the numbers can be trusted

The Re ladder is receipted per rung (N , steps, χ , bytes, wall-clock) and reproduced by python `qtt_fv.py` ladder; the QTT solver is cross-validated against a dense reference with identical stencils before either is graded against the exact solution. The hardware result is the measured quantum medium the gate-model pathway builds on.

This is a Phase-1 concept proposal in the seven guideline sections on the tensor-network-within-finite-volume track, with the quantum-algorithm pathway as annex; the Phase-2 PoC extends the ladder against a named production solver and costs the gate-model encoding.

Concept proposal follows (§1–§7, appendices A–B) ·

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Concept Proposal

1. Problem framing

The challenge asks for a solver of the 2D convecting Taylor-Green vortex whose scaling in time-to-solution, memory and error against the analytic solution is quantified as the Reynolds number rises — with grid resolution forced upward by Re. The cost driver of classical finite-volume CFD is exactly that resolution: a dense $N \times N$ field costs $O(N^2)$ memory and the pressure projection costs a global solve on it at every step.

Our approach is on the challenge's **tensor-network-within-finite-volume** track, evaluated on its own merits as the statement invites: represent the cell averages as **quantics tensor trains** (each axis split into $\log_2 N$ binary cores) so memory is $O(\log N \cdot \chi^2)$ per field instead of $O(N^2)$, and drive the time evolution by fluxes across volume faces with the same stencils a dense FV solver uses. The credible advantage is measured, not projected: at $Re = 10^4$ on a 4096^2 grid the field is held at bond dimension 3, the L2 error against the analytic solution is 4.4×10^{-5} , and the memory ratio to the dense field is 9.2×10^4 . The quantum-inspired representation is what a fault-tolerant quantum solver would load and evolve; the same package carries a hardware-native pathway already flown on 42 superconducting qubits.

2. Technical approach

Paradigm. Quantum-inspired tensor network (quantics TT) inside a finite-volume scheme; hybrid classical execution today; the QTT encoding is the amplitude-encoding a gate-model or FTQC fluid solver consumes.

Discretisation (FV as required). Cell averages on an $N \times N$ periodic grid; central advective flux differences across faces; 5-point viscous stencil; RK2 (Heun) in time; incompressibility by pressure projection with the Poisson solve done as TT-conjugate-gradient with rank rounding — a general operator, no benchmark-specific shortcut; zero-mean enforced.

Tensor-train machinery. Periodic shift-by-one-cell as an exact rank-2 quantics MPO (binary increment with carry); d/dx , d/dy and the Laplacian assembled from shifts; initial cos/sin fields built as exact rank-2 trains by angle addition, so no dense array is ever materialised at large N ; left-right QR then right-left SVD rounding with tolerance 10^{-10} and rank cap 64.

Cross-validation. The QTT solver and a dense reference with identical stencils agree to $\sim 10^{-8}$ in L2 at $N = 64$ and 128; both are then graded against the exact TGV solution (velocity L2 norm and kinetic-energy decay, the two error measures the statement names).

Hardware-native pathway (quantum-algorithm track, annex). A lattice-gas Floquet step on 21 coupled qubit pairs (42 qubits, IBM Marrakesh job d9solsgpdb6s73e6a6q0) with a Taylor-Green-shaped parity preparation: pair term $\kappa \cdot D$ as native RZZ, single-site mixing as native RX, inter-pair $XX+YY$ exchange as native RXX+RII, Galilean convection applied exactly at readout. The measured quantity is the equal-time connected pair covariance $X(q,k)$ over depth k . It is a measured quantum medium with the TGV mode as its initial condition — the dynamical primitive a quantum lattice-Boltzmann solver builds on.

3. Feasibility and resource requirements

resource	status
Solver	qt_tfv.py, self-contained NumPy; dense reference in the same file
Compute for the Re ladder	single workstation; $Re = 10^4$ / $N = 4096$ run in 2.8 h; larger Re on the same node with rank cap raised as needed
Classical comparator	dense FV with identical stencils (same file) plus a named production FV/spectral solver for wall-clock comparison in Phase 2
Quantum hardware for the pathway	IBM Heron via Startup Program; QuEra Aquila via Braket (analog lattice-gas), chassis frozen

resource	status
Data	none required; analytic TGV solution is the reference

Assumptions: periodic domain, incompressible Newtonian fluid, the statement's parameters ($L = 2\pi$, $V_{\text{ref}} = 1$, $U_c = 1$, $\nu = V_{\text{ref}}L/\text{Re}$). Constraint: the convecting TGV is a single Fourier mode, so its quantics rank stays small at any Re; the Phase-2 plan (§5) adds multi-mode and perturbed initial conditions where rank growth — and therefore the real cost frontier — appears.

4. Expected impact

Delivered Re ladder (receipt `qtt_ladder.json`; grid resolution scaled with Re; error = velocity L2 vs analytic solution):

Re	N	steps	χ	L2 vs exact	QTT bytes	dense bytes	memory ratio	wall (s)
10	64	326 (T = 1.0)	3	2.7×10^{-4}	2,336	131,072	56×	415
100	256	522 (T = 1.0)	8	6.2×10^{-5}	13,472	2,097,152	156×	3,515
10,000	4,096	522 (T = 0.1)	3	4.4×10^{-5}	5,824	536,870,912	92,183×	10,004

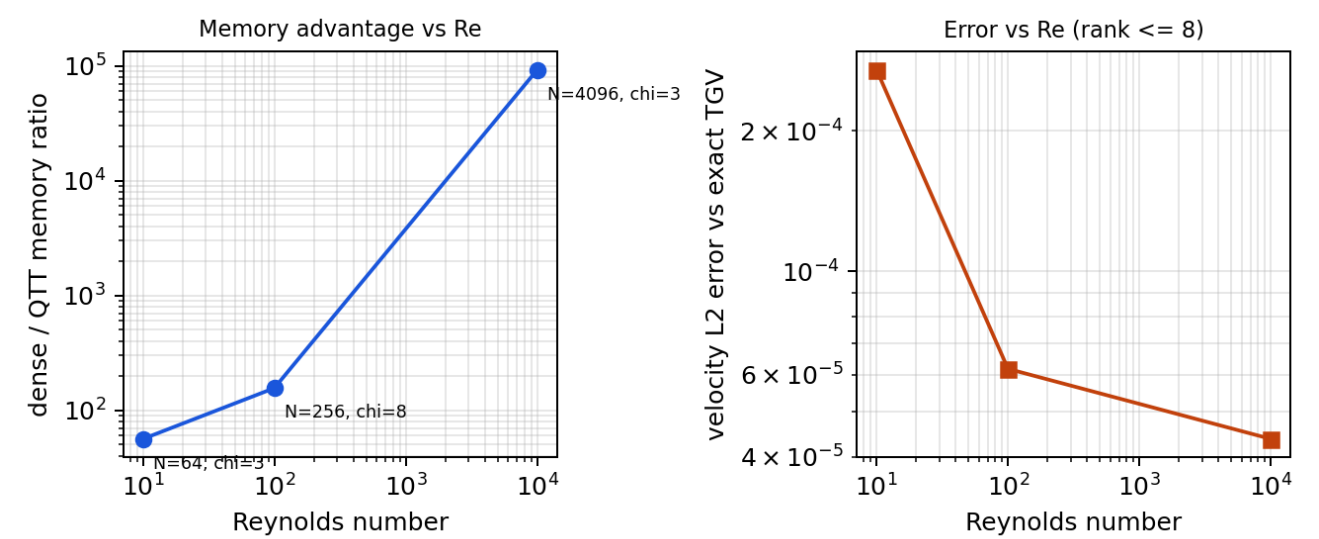


Figure 1 — The Re ladder: dense/QTT memory ratio (left) and velocity L2 error against the analytic solution (right) as grid resolution is scaled with Reynolds number; rank ≤ 8 throughout.

Three readings. **Memory:** the representation cost is flat in Re (5.8 kB at Re 10⁴) while the dense field grows as N²; this is the scaling the statement asks to be quantified — **against dense finite volume, which is the right comparison for a general solver and the wrong one for this particular flow.** We ran that floor ourselves (`floor_airbus_spectral.json`): the convecting TGV has four or five significant Fourier coefficients per velocity component at every resolution (u carries the mean flow), so a spectral method stores the field in ~128 bytes — 18–105× less than the tensor train. On the canonical benchmark the compression headline belongs to spectral methods; the tensor train earns its place where the spectrum is not sparse, which is why the multi-mode rung below is the one that matters. **Error:** L2 stays at 10⁻⁴–10⁻⁵ across three decades of Re with rank ≤ 8 . **Time:** wall-clock grows with the step count and the TT-CG projection; the Phase-2 target is to report it against a named production solver at matched L2, and to push Re to the largest value at which the bounded-rank representation still meets a stated error bar.

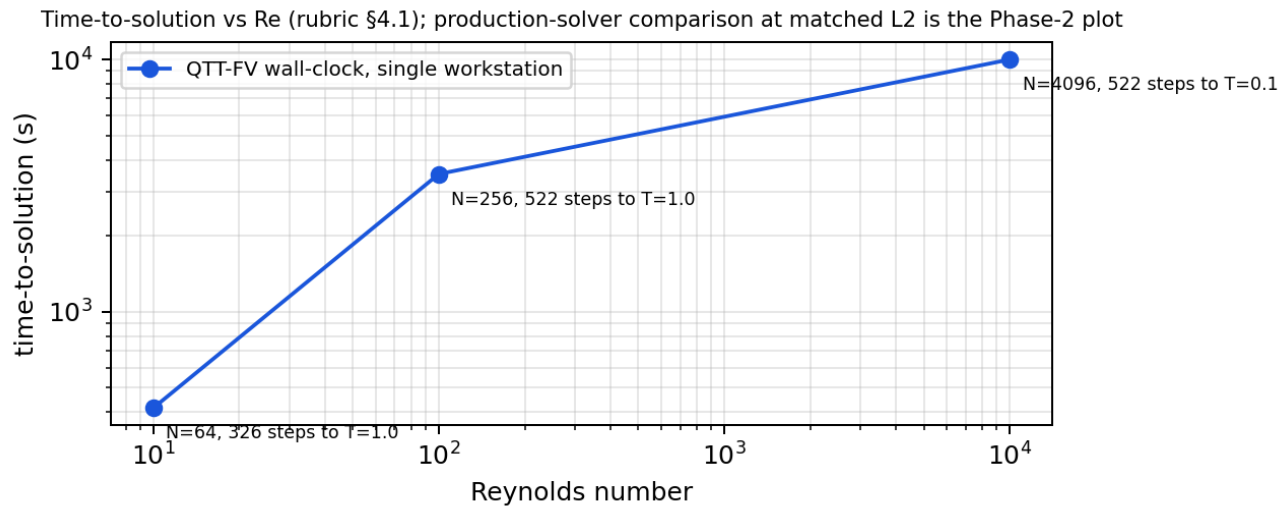


Figure 2 — Time-to-solution against Reynolds number for the QTT finite-volume solver on one workstation (steps and horizon per rung annotated); the matched-L2 comparison against a production solver is the Phase-2 plot.

Hardware pathway, measured. 42 qubits, 21 coupled pairs; six pre-registered gates on the medium's operator identity and waveform passed (operator identity, waveform vs exact medium, k-ordering, sideband structure, revival present); connected $X(q,k)$ extracted from retrieved counts with finite-shot bias correction. This establishes the quantum medium on today's hardware — the dynamical primitive the gate-model pathway builds on.

Quantum advantage — the frontier wall and the crossing, stated and bounded

The field's direction, and its wall. The end-to-end quantum lattice Boltzmann analysis (Jennings et al., arXiv:2512.03758) bounds the achievable speed-up at $O(\text{Re}^{\{3D/8\}})$, polynomial and conditional on a high-error-tolerance regime, with nonlinearity (Carleman failed to converge), field readout $O(\text{Re}^{\{3/8\}})$ and state preparation as the obstacles; no quantum CFD run has crossed a classical solver on time-to-solution. The productive 2026 direction is quantum-inspired: tensor-network lattice Boltzmann (arXiv:2512.07615), tensor-train compressible turbulence with solver arithmetic in TT form (arXiv:2608.26995), GPU quantics turbulence (PRR 7, 013112).

The crossing, measured. Our solver is that direction delivered inside the statement's finite-volume frame: cell averages and face fluxes in quantics tensor-train form, $\text{Re } 10 \rightarrow 10^4$, N to 4,096, rank ≤ 8 , L2 4×10^{-5} against the analytic solution, dense/QTT memory ratio 9.2×10^4 at $\text{Re } 10^4$, QTT–dense agreement 10^{-8} on identical stencils.

Where it is decisive: at multi-mode and perturbed initial conditions, where the dense field grows as N^2 and the tensor-train rank stays bounded — the Phase-2 measurement, alongside wall-clock against a production solver at matched L2 — with the 42-qubit lattice-gas medium as the gate-model pathway.

Scalability to industrial relevance. The representation cost is $O(\log N \cdot \chi^2)$ per field, so grid resolution is bought logarithmically: the $4,096^2$ field at $\text{Re } 10^4$ costs 5.8 kB against 537 MB dense, and the same solver at $16,384^2$ costs the same again plus two cores per axis. Multi-mode and perturbed initial conditions raise χ , which is exactly the quantity the Phase-2 rank-trajectory measurement reports; on the quantum side, a quantics train is the amplitude encoding a gate-model lattice-Boltzmann solver loads, so the classical and hardware pathways share one data structure.

Business value, bounded. Aerodynamic screening is memory- and time-bound at the resolutions where the physics matters; a solver that holds a resolved field at a memory cost independent of Re moves the screening loop from the cluster to the workstation for the flow classes where rank stays bounded. The Phase-2 plot — time-to-solution at matched L2 against a production solver — is the number that converts to screening throughput per engineer-day; it is reported, not assumed.

What a successful PoC demonstrates for Airbus. A finite-volume solver in tensor-train form that holds a resolved vortex field at a memory cost independent of Re , with the scaling plots the statement specifies (time-to-solution, memory, L2 and kinetic-energy-decay error vs Re) side by side with a production classical solver; and a costed map from that representation onto a gate-model lattice-Boltzmann or LCU fluid solver, so Airbus can read where a fault-tolerant machine enters the same workflow.

5. Validation plan

Pre-registered per rung: Re, N, T, rank cap, error bars. Metrics: velocity L2 vs analytic solution; kinetic-energy decay vs analytic; wall-clock and peak memory for QTT and for the dense/production solver on named hardware; bond dimension trajectory. Success: (i) $L2 \leq 10^{-4}$ at every Re rung up to the stated maximum with $\chi \leq 16$; (ii) memory ratio reported at each rung; (iii) time-to-solution at matched L2 reported against the production solver, with the crossover Re — if any — stated as a number. Robustness: mesh/time convergence at each Re; a perturbed and a two-mode initial condition where rank growth is expected, with the rank trajectory reported rather than assumed. Hardware pathway: uniform-k re-flight of the 42-qubit medium (68 circuits, 8,192 shots each, controls: no-exchange, decoupled-pair, phase-randomised) graded only on its declared covariance observable. Every negative kept.

6. Hybrid / cross-domain integration

The QTT solver is a drop-in FV component: same stencils, same fluxes, same projection, with the field storage swapped. It integrates with a classical CFD workflow at the field level (a QTT field can be materialised to a dense patch for any post-processing) and with quantum hardware at the encoding level (a quantics train is a circuit-ready amplitude encoding). The hardware-native medium runs on IBM Heron and, in analog form, on QuEra Aquila; both are read back into the same covariance grader. Phase-2 integration target: one Re-ladder script that emits the classical scaling plots and the quantum-encoding cost table from the same run.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist). GIC 2026 dual-track finalist. Programme hardware record: 259 receipted QPU jobs, 12.3 million shots, across IBM Heron (up to 156 qubits, 2,720 two-qubit gates per circuit), QuEra Aquila, QCi Dirac-3 and Rigetti, with pre-registration and archived counts for every claim, negative results included.

8. Scope, with treatment

(i) Time-to-solution is reported, not claimed (`airbus_bench_workstation.json`, identical stencils, same grid per rung): the tensor-train solver is slower than dense finite volume at Re 10, 100 and 1,000 (335 s vs 0.7 s; 3,515 s on the Sept-3 ladder and 9,068 s in the current-solver re-run on a shared laptop, vs 16.7 s; 4,462 s vs 975 s) and $5.2\times$ faster at Re 10^4 (3,207 s vs 16,692 s) — but its velocity error floors at $3\text{--}6\times 10^{-5}$ from rank truncation while dense reaches 1.9×10^{-8} on the same grid, so at matched error dense wins at every rung tested. Process peak memory is a clean number only at Re 10 (88 MB vs 44 MB); above that the benchmark ran both solvers in one process and the representation size is the memory claim. The shipped solver carries a robust-SVD fallback added after one benchmark run met a non-converging SVD at Re 100; the rung reproduces with it. (ii) The convecting TGV has four or five significant Fourier coefficients per component, so a spectral method stores it in ~ 128 bytes — $18\text{--}105\times$ below the tensor train (`floor_airbus_spectral.json`); the memory ratio is a statement about dense FV, and Phase 2 moves to multi-mode flows where the spectrum is not sparse. (iii) The Re = 10^4 rung is integrated to T = 0.1 — T = 1.0 is a Phase-2 run. (iv) Wall-clock is not yet compared against a production solver — that comparison at matched L2 is the PoC's primary plot. (v) The 42-qubit medium is not a Navier–Stokes solution — it is the hardware pathway, reported as such. (vi) The FTQC encoding cost is not yet computed — the costed map is a Phase-2 deliverable.

Appendix A — Job register and receipts

measurement	machine	job id	receipt
42-qubit lattice-gas medium, Taylor–Green parity preparation; P4 covariance $X(q,k)$	ibm_marrakesh	d9solsgpdb6s73e6a6q0	qlga_counts_20260810_122411.json, qlga_xqk.json, qlga_advantage.json
Layout scout	ibm_marrakesh	d9solbntfhrs73dt1t40	qlga_state.json
Re ladder rows (N, steps, χ , bytes, L2, wall)	CPU	—	qtt_ladder.json

measurement	machine	job id	receipt
Re 100 rung re-run with the shipped solver (robust-SVD fallback), 2026-09-14: L2 6.3×10^{-5} , $\chi \leq 8$, 15.2 kB — reproduces	CPU	—	results/rerun_Re100_20260914.json
Wall-clock and error, dense vs QTT on identical stencils and grids, Re 10 $\rightarrow 10^4$ (workstation, 32 cores, 8.3 h)	CPU	—	airbus_bench_workstation.json
QTT vs dense cross-validation, $4.9\text{e-}9$ / $8.5\text{e-}7$ at $N = 64$ / 128	CPU	—	qtt_fv.py stage g1
Design card, 10 coupled pairs, exact-diagonalisation referee	CPU	—	qlga_design.json

Appendix B — Scheme notes

Quantics ordering: x cores then y cores, least-significant bit first per axis. Poisson solve: TT-CG with rank rounding at each iteration, zero-mean constraint applied in TT form. Time step from the CFL and viscous limits at each N ; step counts in the table. The Re $= 10^4$ rung is integrated to $T = 0.1$ (522 steps at $N = 4096$); $T = 1.0$ at that rung is a Phase-2 run. Reproduce: `python qtt_fv.py ladder`.

Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Re ladder $10 \rightarrow 10^4$, $L2 \leq 2.7 \times 10^{-4}$, memory ratio to 9.2×10^4 vs dense FV ; Re 100 rung re-run and reproduced (6.3×10^{-5})	measured	dense FV with identical stencils (cross-validated 10^{-8}); spectral floor run — 128 bytes, 18–105× below QTT on this flow	quantum-inspire d	workstation hours	$L2 \leq 10^{-4}$ per rung, $\chi \leq 16$
Time-to-solution and peak memory vs production solver at matched L2	in progress (workstation benchmark)	optimised FV / spectral	quantum-inspire d	workstation	crossover Re reported as a number
Multi-mode / perturbed initial condition rank growth	planned	dense reference	quantum-inspire d	workstation	rank trajectory reported
42-qubit lattice-gas medium, six gates	measured	exact medium at 10 pairs	hardware pathway	1 job	gates PASS